

Disproof of conjecture 00000002051

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2026-09-13

Verdict: FALSE

We disprove conjecture 00000002051 as stated. The conjecture claims that the fine spectrum of the semigroup variety $\langle x^2 = x \rangle$ is the closed form $f(n) = 2^n C_n$. The variety $\langle x^2 = x \rangle$ is the variety of *bands* (idempotent semigroups), and the fine spectrum is by definition the cardinality of the n -generated free algebra. The formula fails already at $n = 1$: the free band on one generator is a singleton, so $f(1) = 1$, whereas the formula gives $2^1 C_1 = 2$. An independent second failure at $n = 2$ ($f(2) = 6 \neq 8$) shows the claim cannot be repaired by restricting or shifting the range of n .

1 The conjecture and its terms

Quoted verbatim from `conjectures/00000002051.md`:

English. Definition: The fine spectrum $f_V(n)$ of a variety V is the cardinality of its n -generated free algebra. Conjecture: The fine spectrum of the semigroup variety $\langle x^2 = x \rangle$ is the explicit closed form $f(n) = 2^n \cdot C_n$ (C_n the Catalan numbers); the spectrum is controlled by the connected-component count of band semigroups.

The file fixes the meaning of its two key terms itself:

Definition 1 (fine spectrum). $f_V(n) = |F_V(n)|$, the number of elements of the free algebra of the variety V on n generators.

Definition 2 (the variety $\langle x^2 = x \rangle$). A semigroup in which $x^2 = x$ for every element x is an *idempotent semigroup*, i.e. a *band*. Thus $F_V(n)$ is the free band on n generators, and $f(n)$ is its cardinality. The file's own concluding clause ("band semigroups") confirms this reading.

The natural range is $n \geq 1$. The claim is a single closed form asserted for this range; a single value of n at which the two sides differ refutes it.

2 Failure at $n = 1$

Theorem 3. *The free band on one generator has exactly one element; that is, $f(1) = 1$. The conjecture's formula gives $2^1 C_1 = 2 \neq 1$.*

Proof. Let B be the free band on the single generator x , and let $S \subseteq B$ be the subsemigroup generated by x . Every element of S is a finite product $xx \cdots x$. Since B is a band, $x \cdot x = x$; multiplying a product of k copies of x by one more copy gives $x^k \cdot x = x^{k+1}$, and induction on k

using $x \cdot x = x$ shows $x^k = x$ for every $k \geq 1$. Hence every element of S is x , so $S = \{x\}$ is a singleton and $B = S$. Therefore $|B| = f(1) = 1$.

More generally, in *any* band B and for any element $x \in B$, the subsemigroup generated by x is the singleton $\{x\}$: the set $\{x\}$ is closed under multiplication precisely because $x \cdot x = x$. This universal statement is what makes the refutation independent of any structural theory of bands. The free band on one generator is unique up to isomorphism and has one element, so

$$f(1) = 1 \neq 2 = 2^1 C_1.$$

□

Corollary 4. *Conjecture 00000002051 is false. The closed form $f(n) = 2^n C_n$ already disagrees with the fine spectrum at the smallest admissible value $n = 1$.*

Remark 5. The argument is a single line: $2^1 C_1 = 2$ but the free band on one generator is $\{x\}$. No generating set of size 1 can produce two distinct elements, because the only word one can form is x itself.

3 Failure at $n = 2$

The second failure is independent of Theorem 3.

Theorem 6. *The free band on two generators has exactly six elements*

$$\{a, b, ab, ba, aba, bab\}.$$

Hence $f(2) = 6$, whereas the conjecture's formula gives $2^2 C_2 = 4 \cdot 2 = 8$.

Proof. Let a, b be the free generators. The free band is the quotient of the free semigroup $\{a, b\}^+$ by the congruence generated by $u \cdot u = u$ for all words u (idempotency of every element). We first observe that *every word of length ≥ 4 over a two-letter alphabet contains a square factor*, a factor ss consisting of two equal adjacent blocks. Indeed it suffices to check the 16 words of length 4 over $\{a, b\}$; each contains a square factor (among them $aa, bb, abab, abba, baab, baba$), and every word of length ≥ 4 has its first four letters as a factor. Consequently any word with a factor ss reduces, by $s \cdot s = s$, to the strictly shorter word with one copy of s removed; iterating, every word reduces to a *square-free* word of length ≤ 3 . The square-free words of length ≤ 3 over $\{a, b\}$ are exactly the six displayed words (length 1: a, b ; length 2: ab, ba ; length 3: aba, bab ; the words $aa, bb, aab, abb, aaa, bbb, \dots$ all contain a square). Therefore $f(2) \leq 6$.

For the reverse inequality, the induced multiplication on the six reduced words (taking reduced concatenation) is the table below; the entry in row u and column v is the reduced form of the concatenation uv . □

$\cdot$	a	b	ab	ba	aba	bab
a	a	ab	ab	aba	aba	ab
b	ba	b	bab	ba	ba	bab
ab	aba	ab	ab	aba	aba	ab
ba	ba	bab	bab	ba	ba	bab
aba	aba	ab	ab	aba	aba	ab
bab	ba	bab	bab	ba	ba	bab

Proposition 7. *The table is closed, associative, and idempotent.*

Proof. Closure is immediate from the table (all entries lie in the six-word set). Associativity is the assertion $u(vw) = (uv)w$ for all $6^3 = 216$ triples; idempotence is $u^2 = u$ for the six diagonal entries. Both were verified by exhaustive computation in `reproduce.py`, which reduces all words over $\{a, b\}$ up to length 8 and checks the table entry by entry. The six diagonal entries are a, b, ab, ba, aba, bab , so the checks pass. $\square$

Corollary 8. *The six-element table is a band. It is generated by the two elements a and b , so by the universal property of the free band it is a quotient of the free band on two generators, giving $f(2) \geq 6$. Combined with $f(2) \leq 6$ from the reduction argument, $f(2) = 6$. The conjecture’s value is 8, so it fails at $n = 2$ as well.*

Remark 9. Six is a classical value: the free band on two generators has six elements (Howie, *Fundamentals of Semigroup Theory*, Oxford University Press, 1995, p. 123). It is the second term of OEIS A030449, whose terms are 1, 6, 159, 332380, $\dots$

4 True free-band values versus the conjectured formula

The number of elements of the free band on n generators is OEIS **A030449** (Howie 1995, p. 123). The values differ from $2^n C_n$ for every $n \geq 1$, and the discrepancy grows:

n	free band $f(n)$ (A030449)	$2^n C_n$	equal?
1	1	2	no
2	6	8	no
3	159	40	no
4	332380	224	no

Here $C_1, C_2, C_3, C_4 = 1, 2, 5, 14$. The formula overcounts at $n = 1, 2$ and drastically undercounts from $n = 3$ on: $2^n C_n$ is a genuine sequence 2, 8, 40, 224, $\dots$, but it is not the band fine spectrum.

5 Alternative readings

The refutation is robust under the readings a defender of the conjecture might propose.

1. “ $\langle x^2 = x \rangle$ means a different variety or a free object $\langle x \mid x^2 = x \rangle$.” A semigroup with $x^2 = x$ is an idempotent semigroup, i.e. a band, and the file says “band semigroups”. The presentation $\langle x \mid x^2 = x \rangle$ has a single generator x ; its free object on one generator is still $\{x\}$. Both readings agree and both give $f(1) = 1 \neq 2$.
2. “ $f(n)$ counts irreducible or maximal words rather than algebra elements.” The definition says “cardinality of its n -generated free algebra”. Even under this alternative reading the counts for $n = 1, 2$ are the irreducible-word counts 1 and 6, hence still differ from 2 and 8.
3. “ n starts at 2.” Then $f(2) = 6 \neq 8 = 2^2 C_2$; the claim is still false. Shifting further does not help: $f(3) = 159 \neq 40$ and $f(4) = 332380 \neq 224$, and the two sequences never agree for $n \geq 1$.
4. “Adjoin an identity to the free band.” Under this non-standard reading $f(1) = 2 = 2^1 C_1$ happens to match, but $f(2) = 6 + 1 = 7 \neq 8 = 2^2 C_2$, so the claim still fails. (The conjecture

speaks of a *semigroup* variety, where no identity is adjoined, so this is not the intended reading in any case.)

5. $n = 0$. The free semigroup, hence free band, on the empty set is empty, so $f(0) = 0$, whereas the formula gives $2^0 C_0 = 1$. If one instead declares the 0-generated free band to be trivial, then $n = 0$ happens to match, but the failures at $n = 1$ and $n = 2$ are unaffected.

No reading we could construct saves the claim. The refutation is unconditional, and the cheapest formalisable witness is $n = 1$: since the formula is asserted as a closed form for all $n \geq 1$, the single value $f(1) = 1 \neq 2 = 2^1 C_1$ already suffices.

6 Reproducibility and formalisation

The computations of Sections 4 and 5 are reproduced by `reproduce.py`, which uses the Python 3 standard library only. It enumerates all words over $\{x\}$ and over $\{a, b\}$ up to length 8, reduces them by $u \cdot u \rightarrow u$ to a fixed point (collapse the longest square factor first), and asserts

- for $n = 1$: exactly one reduced class, $\{x\}$, so $f(1) = 1 \neq 2 = 2^1 C_1$;
- for $n = 2$: exactly six reduced classes $\{a, b, ab, ba, aba, bab\}$, with a multiplication table that is closed, associative (all 216 triples) and idempotent, and generated by $\{a, b\}$, so $f(2) = 6 \neq 8 = 2^2 C_2$;
- every word of length 4 through 12 over two letters has a square factor (the mechanism that makes the six words complete);
- the literature values 1, 6, 159, 332380 (A030449) differ from $2^n C_n = 2, 8, 40, 224$ for $n = 1, 2, 3, 4$;
- the alternative readings of Section 6 also fail.

It prints **PASS** and exits 0 exactly when every check holds, and prints **FAIL** with a non-zero exit status otherwise.

The accompanying Lean 4 project under `lean4/` formalises the core of the disproof in core Lean only (`import Std`, no `Mathlib`, no `sorry`). It defines the structure of a **Band** (an associative, idempotent binary operation) and proves that in *any* band the subsemigroup generated by one element is the singleton $\{x\}$; that the one-element band has the universal property of the free band on one generator, so $f(1) = 1$; that the explicit six-element multiplication table on **Fin 6** is associative and idempotent by `decide` and is generated by two elements; that the Catalan numbers satisfy $C_1 = 1$, $C_2 = 2$, $C_3 = 5$; and the arithmetic failures $1 \neq 2^1 C_1$, $6 \neq 2^2 C_2$, $159 \neq 2^3 C_3$. The axiom audit (`lake env lean Check.lean`) reports no `sorryAx` and no `Mathlib` dependency. See `lean4/README.md` for the full statement table and the scope note.

Files and rule-3 status

Rule 3 requires the LaTeX source, a PDF, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article compiling with `tectonic main.tex`.
- **PDF** — `build/main.pdf`, produced by `tectonic` and placed in `build/`.

- **Lean 4 project** — `lean4/` with `lean-toolchain (leanprover/lean4:v4.33.1)`, `lakefile.toml` (`library Main, name tlmc2051`), `Main.lean`, `Check.lean`, and `README.md`.

`reproduce.py` accompanies the submission and certifies the finite computations.